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HPC Optimisation Senior Scientific Software Engineer

Met Office

Apply before 11:55 pm on Sunday 22nd November 2026

Reference number

489677

Salary

£44,589 - £48,363

Your total reward package will be up to £65,024 annually, which includes a Recruitment Retention Allowance (RRA) you will be paid £2,700 per annum as a market supplement to reflect the demand for your skills.

A Civil Service Pension with an employer contribution of 28.97%

Job grade

Senior Executive Officer

Contract type

Permanent

Type of role

Engineering
Science

Working pattern

Flexible working, Full-time

Number of jobs available

1

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Location

Exeter, South West England, EX1 3PB

About the job

Job summary

We re looking for an exceptional **HPC Optimisation Senior Scientific Software Engineer** to help us make a difference to our planet.

As our HPC Optimisation Senior Scientific Software Engineer, the job may be suitable for hybrid working, which is where an employee works part of the week in the office and part of the week from home. This is a voluntary, non-contractual arrangement and the location advertised will be your contractual place of work.

Our opportunity is full time, 37 hours per week. Our people are at the heart of what we do and we'll do our best to agree a working pattern that works for everyone.

World changing work

From science to technology, from meteorology to management, and from planning to communication, our expertise helps us stand out as the authority on weather accuracy and climate prediction. We help individuals, industries and government to make better decisions to stay safe and thrive. This is the Met Office. This is who we are.

- We re a force for good - focusing on our environmental and social impact
- We re experts by nature - always learning and developing to do things better
- We live and breathe it - putting our purpose at the heart of decision-making
- We re better together - understanding partnerships and inclusivity make us greater
- We keep evolving - pushing boundaries to make tomorrow better for our customers

Your world of expertise

As a HPC Optimisation Senior Scientific Software Engineer you will contribute to the development, support, and optimisation of large-scale parallel climate and weather applications on HPC systems.

We are currently in a major push to develop our Next Generation Modelling Systems, to help exploit our new world-leading supercomputer capability and to support our adoption of data science (specifically artificial intelligence and machine learning) across our national capability, and you will be working towards these objectives in some form.

The Senior Scientific Software Engineer (SSE) role is a varied one, with many opportunities for development, progression and finding new challenges both within the SSE career family and beyond.

Being a Senior SSE is about far more than just technical skills you will need to be able to work alongside colleagues in a range of areas, both technical and scientific. We frequently require our SSEs to collaborate internally and internationally.

Job description

Analyse and optimise the performance of complex, large-scale scientific applications.

Design and conduct benchmarks to assess the performance of scientific applications on current and future supercomputer hardware.

Analyse and tune the performance of operational scientific workloads, working with the teams responsible for those workloads to identify and address performance bottlenecks.

Apply HPC expertise to diagnose and resolve complex performance and technical issues, providing guidance to users and colleagues.

Develop novel software, tools and methods to improve the performance, scalability and portability of large-scale scientific applications on current and future HPC architectures.

Collaborate with and provide technical expertise to scientific and technical colleagues in the development and improvement of HPC applications.

Maintain, extend and share expertise in optimisation tools, parallel programming techniques and algorithms, and current and future supercomputer capabilities.

Person specification

1. **Lead Criteria:** Demonstrate insight in High-Performance Computing Code Optimisation through improvements in computational performance of large-scale scientific applications and systematic use of code performance analysis tools. (We re Experts by Nature)
2. Engage effectively with internal and external stakeholders to understand requirements and the wider context of your work. Contribute significantly to developing plans that meet these requirements and inform wider strategic planning. (We re Experts by Nature, Better Together)
3. Promote good Quality Assurance processes, best practice, standards and/or regulations in your work and that of others. (We re Experts by Nature, We Live and Breathe it)
4. Communicate knowledge accurately and concisely through written communication and discussions, including with larger groups, adapting your approach for diverse audiences and building trust and credibility. (We re Better Together)
5. Proactively develop your own technical skills and support others to develop theirs through activities such as knowledge sharing, mentoring, coaching or collaboration, contributing to improved individual, team or organisational outcomes. (We Keep Evolving, Better Together, Experts by Nature)

The panel may perform a preliminary sift of the lead criteria as indicated above.

Benefits

Alongside your salary of £44,589, Met Office contributes £12,917 towards you being a member of the Civil Service Defined Benefit Pension scheme. Find out what benefits a Civil Service Pension provides (opens in a new window).

Our work is life-changing, often life-saving and always life-enhancing. The Met Office is Great Place to Work UK certified. We are also featured on their Best

Workplaces in Tech 2023, 2024 and 2025 lists, as well as their 54 Best Workplaces for Women 2023 list.

As our HPC Optimisation Senior Scientific Software Engineer, your total reward package will be up to £65,024 annually, which includes:

- £44,589 base pay
- An outstanding Civil Service pension, with an average employer contribution of 28.97%
- Recruitment Retention Allowance (RRA) you will be paid £2,700 per annum as a market supplement to reflect the demand for your skills. Whilst in post, you will be paid this market driven allowance from April 2026 until March 2027 in your monthly pay
- Annual Leave starting at 27.5 days (plus Bank Holidays) rising to 32.5 days (plus Bank Holidays) after 5 years and option to buy or sell up to 5 days per year of annual leave

Things you need to know

Artificial intelligence

Artificial intelligence can be a useful tool to support your application, however, all examples and statements provided must be truthful, factually accurate and taken directly from your own experience. Where plagiarism has been identified (presenting the ideas and experiences of others, or generated by artificial intelligence, as your own) applications may be withdrawn and internal candidates may be subject to disciplinary action. Please see our candidate guidance (opens in a new window) for more information on appropriate and inappropriate use.

Selection process details

If you share our values, we d love to hear from you! Click apply to begin your application. Please complete your career history and provide evidence against each of the essential criteria in the supporting statement questionnaire. We recommend candidates use the CARL method (Context, Action, Result and Learning) for presenting evidence of experience and skills.

Please note: We are currently recruiting for 1 role which is being advertised at 2 levels (Scientific Software Engineer and Senior Scientific Software Engineer). Those wishing to be considered at multiple levels should submit an application at each level.

Closing date 22/11/2026 at 23:59 with first stage interviews commencing from 07/12/2026. You will hear from us once the closing date has passed.

Using AI in your application

We welcome applications that use AI tools for support in drafting or refining, as long as they accurately reflect your own skills and experience. All hiring decisions at the Met Office are made by people, not AI. For more details, visit our approach to recruitment.

How we can help

If you have any questions or would like to discuss this opportunity further, please contact us at careers@metoffice.gov.uk.

If you re considering applying and need support to do so, please get in touch. You can request adjustments either within your application or by contacting us. Should you be offered an interview, please be aware there may be a selection exercise which could include a presentation, written test or a scenario-based activity. You can select in your application to be considered under the Disability Confident Scheme. To be invited to interview/assessment under this scheme, your application must meet the essential criteria for the role.

We understand that great minds don t always think alike and as an equal opportunities employer we welcome applications from those with all protected characteristics. We recruit on merit, fairness, and open competition in line with the Civil Service Code.

We can only accept applications from those eligible to live and work in the UK - please refer to GOV.UK for information. We require Security clearance without caveats and/or restrictions for this role, for which you need to have resided in the UK for at least 3 of the last 5 years to be eligible, 2 of these years must be immediately preceding the point of your application.

If you feel that your application has not been treated in accordance with the Recruitment Principles, and wish to make a complaint, then in the first instance you should contact us at: **careers@metoffice.gov.uk** If you are not satisfied with the response that you receive, then you can contact the Civil Service Commission at: **info@csc.gov.uk**

Feedback will only be provided if you attend an interview or assessment.

Security

Successful candidates must meet the security requirements before they can be appointed. The level of security needed is security check (opens in a new window).

See our vetting charter (opens in a new window).

People working with government assets must complete baseline personnel security standard (opens in new window) checks.

Medical

Successful candidates will be expected to have a medical.

Nationality requirements

This job is broadly open to the following groups:

- UK nationals
- nationals of the Republic of Ireland
- nationals of Commonwealth countries who have the right to work in the UK
- nationals of the EU, Switzerland, Norway, Iceland or Liechtenstein and family members of those nationalities with settled or pre-settled status under the European Union Settlement Scheme (EUSS) (opens in a new window)
- nationals of the EU, Switzerland, Norway, Iceland or Liechtenstein and family members of those nationalities who have made a valid application for settled or pre-settled status under the European Union Settlement Scheme (EUSS)

individuals with limited leave to remain or indefinite leave to remain who were eligible to apply for EUSS on or before 31 December 2020

Turkish nationals, and certain family members of Turkish nationals, who have accrued the right to work in the Civil Service

Further information on nationality requirements (opens in a new window)

Working for the Civil Service

The Civil Service Code (opens in a new window) sets out the standards of behaviour expected of civil servants.

We recruit by merit on the basis of fair and open competition, as outlined in the Civil Service Commission's recruitment principles (opens in a new window).

The Civil Service embraces diversity and promotes equal opportunities. As such, we run a Disability Confident Scheme (DCS) for candidates with disabilities who meet the minimum selection criteria.

Diversity and Inclusion

The Civil Service is committed to attract, retain and invest in talent wherever it is found. To learn more please see the Civil Service People Plan (opens in a new window) and the Civil Service Diversity and Inclusion Strategy (opens in a new window).

Apply and further information

The Civil Service welcomes applications from people who have recently left prison or have an unspent conviction. Read more about prison leaver recruitment (opens in new window).

Once this job has closed, the job advert will no longer be available. You may want to save a copy for your records.

Contact point for applicants

Job contact :

Name : Resourcing Team

Email : careers@metoffice.gov.uk

Recruitment team

Email : careers@metoffice.gov.uk

Further information

careers@metoffice.gov.uk

Civil Service Commission

disability confident EMPLOYER